1. Given the following system of equations:

$$\begin{array}{ccccccc} x & - & 3y & + & z & = & 1 \\ x & + & y & + & 2z & = & 14 \end{array}$$

find all the solutions using Gauss-Jordan elimination procedure. Is this an example of consistent system? Why?

2. Show that the following linear system:

$$\begin{array}{ccccccccccc} x_1 & - & x_2 & & & & & & & = & b_1 \\ & & x_2 & - & x_3 & & & & & = & b_2 \\ & & & & x_3 & - & x_4 & & & = & b_3, \\ & & & & & & x_4 & - & x_5 & = & b_4 \\ -x_1 & & & & & & & + & x_5 & = & b_5 \end{array}$$

has solution if and only if $\sum_{i=1}^{5} b_i = 0$.

3. In a certain sense, the following systems is not linear:

$$\begin{array}{ccccccc} 2\sin\alpha & - & \cos\beta & + & 3\tan\gamma & = & 3 \\ 4\sin\alpha & + & 2\cos\beta & - & 2\tan\gamma & = & 10, \\ 6\sin\alpha & - & 3\cos\beta & + & \tan\gamma & = & 9. \end{array}$$

However, there is a away to do Gauss-Jordan elimination on it. Does a solution exist for α, β, and γ?

4. State whether the following statements are true or false. If true explain your answer, if false give an example for which the statement is false:

   (a) If A is a 2×2 such that $A^2 = A \cdot A = 0$, then $A = 0$ (the zero matrix).

   (b) If A, B and C are invertible matrices, then $(A + B)C$ is invertible.

   (c) If A and B are identical except in the upper-left corner, where $b_{11} = 2a_{11}$, then $\det(A) = 2\det(B)$.

   (d) The determinant of a matrix is the product of the pivots.

   (e) If A is invertible and B is singular, then $A + B$ is invertible.

   (f) If A is invertible and B is singular, then AB is singular.